

Solution to Tutorial 2 problems

1.

$$r = k(1 + \cos \theta) \rightarrow (1)$$

$$\dot{r} = -k \dot{\theta} \sin \theta \rightarrow (2)$$

$$\ddot{r} = -k [\dot{\theta}^2 \cos \theta + \ddot{\theta} \sin \theta] \rightarrow (3)$$

Now, $\vec{v} = \hat{r} \dot{r} + \hat{\theta} r \dot{\theta} \rightarrow (4)$

$$\begin{aligned} v^2 &= \dot{r}^2 + r^2 \dot{\theta}^2 \\ &= k^2 \dot{\theta}^2 \sin^2 \theta + k^2 (1 + \cos^2 \theta + 2 \cos \theta) \dot{\theta}^2 \end{aligned}$$

$$= 2k^2 \dot{\theta}^2 (1 + \cos \theta)$$

Thus, $\dot{\theta} = \left[\frac{v^2}{2k^2 (1 + \cos \theta)} \right]^{1/2} \rightarrow (5)$

$$\Rightarrow \boxed{\dot{\theta} = \frac{v}{\sqrt{2kr}}} \quad (\text{using (1)}) \rightarrow (6)$$

Differentiating (5) and using (2) we get:

$$\ddot{\theta} = \frac{v^2 \sin \theta}{4r^2} = \frac{v^2 \sin \theta}{4k^2 (1 + \cos \theta)^2}$$

Now,

$$\vec{a} = (\ddot{r} - r \dot{\theta}^2) \hat{r} + (r \ddot{\theta} + 2 \dot{r} \dot{\theta}) \hat{\theta}$$

So,

$$\begin{aligned} (a) \quad \vec{a} \cdot \hat{r} &= \ddot{r} - r \dot{\theta}^2 \\ &= -\kappa (\dot{\theta}^2 \cos \theta + \ddot{\theta} \sin \theta) \\ &\quad - \kappa (1 + \cos \theta) \dot{\theta}^2 \end{aligned}$$

— doing the simple algebra —

$$\vec{a} \cdot \hat{r} = -\frac{3}{2} \kappa \dot{\theta}^2 (1 + \cos \theta)$$

$$\Rightarrow \boxed{\vec{a} \cdot \hat{r} = -\frac{3}{4} \frac{v^2}{\kappa}}$$

$$(b) \quad \vec{a} \cdot \hat{\theta} = r \ddot{\theta} + 2 \dot{r} \dot{\theta}$$

$$\Rightarrow \boxed{\vec{a} \cdot \hat{\theta} = -\frac{3}{4} \frac{v^2}{\kappa} \frac{\sin \theta}{1 + \cos \theta}}$$

(after doing straightforward algebra!)

$$(c) \quad |\vec{a}| = \sqrt{(\vec{a} \cdot \hat{r})^2 + (\vec{a} \cdot \hat{\theta})^2}$$

$$\Rightarrow \boxed{|\vec{a}| = \frac{3}{4} \frac{v^2}{\kappa} \sqrt{\frac{2}{1 + \cos \theta}}}$$

2.

$$r = A e^{\omega t}, \quad \theta = \omega t$$

$$\vec{v} = \dot{r} \hat{r} + (r \dot{\theta}) \hat{\theta}$$

$$= (\omega A e^{\omega t}) \hat{r} + (\omega A e^{\omega t}) \hat{\theta}$$

$$\vec{a} = (\ddot{r} - r \dot{\theta}^2) \hat{r} + (r \ddot{\theta} + 2 \dot{r} \dot{\theta}) \hat{\theta}$$

$$= (\omega^2 A e^{\omega t} - \omega^2 A e^{\omega t}) \hat{r} + (0 + 2 \omega^2 A e^{\omega t}) \hat{\theta}$$

$$= 2 \omega^2 A e^{\omega t} \hat{\theta}$$

$$|\vec{v}| = \sqrt{2} \omega A e^{\omega t}$$

$$|\vec{a}| = 2 \omega^2 A e^{\omega t}$$

The angle between $\vec{v}$ and $\vec{a}$ is then given by:

$$\begin{aligned} \cos \alpha &= \frac{\vec{v} \cdot \vec{a}}{|\vec{v}| |\vec{a}|} = \frac{2 \omega^3 A^2 e^{2\omega t}}{(\sqrt{2} \omega A e^{\omega t})(2 \omega^2 A e^{\omega t})} \\ &= \frac{1}{\sqrt{2}} \Rightarrow \alpha = \frac{\pi}{4} // \end{aligned}$$

3.

$$(a) \quad r = A\theta = \frac{\theta}{\pi}$$

$$\theta = \alpha t^2/2$$

$$\vec{v} = \dot{r} \hat{r} + r \dot{\theta} \hat{\theta}$$

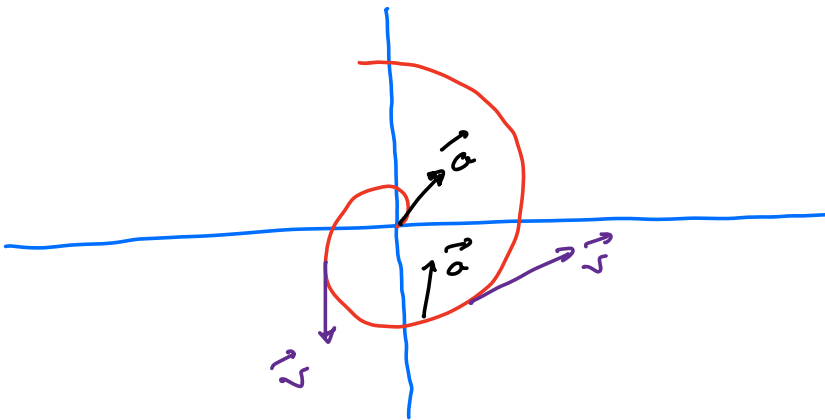
$$\vec{a} = (\ddot{r} - r \dot{\theta}^2) \hat{r} + (r \ddot{\theta} + 2\dot{r} \dot{\theta}) \hat{\theta}$$

$$r = \frac{\alpha t^2}{2\pi} \quad ; \quad \dot{r} = \frac{\alpha t}{\pi} \quad ; \quad \dot{\theta} = \alpha t$$

$$\ddot{r} = \frac{\alpha}{\pi} \quad , \quad \ddot{\theta} = \alpha$$

$$\text{Thus,} \quad \vec{v} = \frac{\alpha t}{\pi} \hat{r} + \frac{\alpha^2 t^3}{2\pi} \hat{\theta}$$

$$\vec{a} = \left(\frac{\alpha}{\pi} - \frac{\alpha^3 t^4}{2\pi} \right) \hat{r} + \left(\frac{5\alpha^2 t^2}{2\pi} \right) \hat{\theta}$$



(b)

radial acceleration

$$a_r = \frac{\alpha}{\pi} - \frac{\alpha^3 t^4}{2\pi} = 0,$$

say at time $t = T$

$$\text{So, } \frac{\alpha}{\pi} - \frac{\alpha^3 T^4}{2\pi} = 0$$

$$\Rightarrow T^4 = \frac{2}{\alpha^2}$$

$$\Rightarrow T^2 = \frac{\sqrt{2}}{\alpha}$$

$$\text{Now, } \theta(t) = \frac{\alpha t^2}{2}$$

$$\text{Thus, } \theta(T) = \frac{\alpha T^2}{2} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \text{ rad}$$

(c)

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2) \hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta}) \hat{\theta}$$

using the results from part (a) we can write:

$$\vec{a} = \left(\frac{\alpha}{\pi}\right) \left[(1 - 2\theta^2) \hat{r} + 5\theta \hat{\theta} \right]$$

$$\Rightarrow a_r = \left(\frac{\alpha}{\pi}\right) (1 - 2\theta^2); \quad a_\theta = \left(\frac{\alpha}{\pi}\right) 5\theta$$

$$|a_n| = |a_o|$$

$$\Rightarrow |1 - 2\theta^2| = |5\theta|$$

If $\theta < \frac{1}{\sqrt{2}}$, then

$$1 - 2\theta^2 = 5\theta$$

$$\Rightarrow 2\theta^2 + 5\theta - 1 = 0$$

$$\Rightarrow \theta = \frac{-5 \pm \sqrt{33}}{4}$$

Because $\theta \geq 0$, $\theta = \frac{-5 + \sqrt{33}}{4}$
 $\approx 0.186 \text{ rad}$
 $\approx 10^\circ$

If $\theta > \frac{1}{\sqrt{2}}$, then

$$2\theta^2 - 1 = 5\theta$$

$$\Rightarrow 2\theta^2 - 5\theta - 1 = 0$$

$$\Rightarrow \theta = \frac{5 + \sqrt{33}}{4} \approx 2.69 \text{ rad}$$

$$= 154^\circ$$

So, the angles (θ) at which the radial and tangential acceleration have equal magnitude, in the given problem, are: $10^\circ, 154^\circ$
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4.

There are two forces on the rubber disc: the net normal force and the force of gravity.

$$\vec{F} = N \hat{r} - mg \hat{z}$$

Since the rubber disc is confined between the cylinders,

$$r = R, \text{ a constant}$$

Now,

$$F_r = m(\ddot{r} - r\dot{\theta}^2)$$

$$F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$$

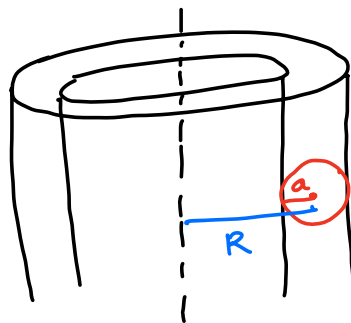
$$F_z = m\ddot{z}$$

$\Downarrow$

$$N = -mR\dot{\theta}^2 \rightarrow (1)$$

$$0 = mR\ddot{\theta} \rightarrow (2)$$

$$-mg = m\ddot{z} \rightarrow (3)$$



$$(2) \Rightarrow \dot{\theta} = \omega, \text{ a constant}$$

$$\Rightarrow \boxed{\theta = \theta_0 + \omega t} \rightarrow (4)$$

The rubber disc moves around the cylinder at a constant rate ω .

$$(3) \Rightarrow \ddot{z} = -g$$

$$\Rightarrow \boxed{z = z_0 + v_{0z}t - \frac{1}{2}gt^2} \rightarrow (5)$$

From (4) and (5) we conclude that the resulting path is a helix of downward increasing pitch.

5.

$$\vec{v} = \hat{r} \dot{r} + \hat{\theta} r \dot{\theta}$$

$$= \frac{2b}{\tau^2} (\tau - t) \hat{r} + \frac{bt}{\tau^3} (2\tau - t) \hat{\theta}$$

$$v^2 = \frac{b^2}{\tau^6} \left[t^4 - 4\tau t^3 + 8\tau^2 t^2 - 8\tau^3 t + 4\tau^4 \right]$$

$$\frac{dv^2}{dt} = \frac{b^2}{\tau^6} \left[4t^3 - 12\tau t^2 + 16\tau^2 t - 8\tau^3 \right]$$

$$= \frac{4b^2}{\tau^6} (t - \tau) \underbrace{(t^2 - 2\tau t + 2\tau^2)}_{\downarrow}$$

always positive

So, $\frac{d}{dt} v^2 \begin{cases} < 0 & \text{for } t < \tau \\ = 0 & \text{for } t = \tau \\ > 0 & \text{for } t > \tau \end{cases}$

Hence, v achieves its minimum value when $t = \tau$. At this instant

$$v = \frac{b}{\tau}$$

Acceleration:

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2) \hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta}) \hat{\theta}$$

$$= \left(-\frac{2b}{\tau^2} - \frac{bt}{\tau^4} (2\tau - t) \right) \hat{r} + \left(0 + \frac{4b}{\tau^3} (\tau - t) \right) \hat{\theta}$$

$$= -\frac{3b}{\tau^2} \hat{r}, \text{ at } t = \tau.$$

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6.

(a)

$$\begin{aligned}\vec{L} &= \vec{r} \times \vec{p} \\ &= m \vec{r} \times \vec{v} \\ &= m \hat{r} r \times (\dot{r} \hat{r} + r \dot{\theta} \hat{\theta}) \\ &= m r^2 \dot{\theta} \hat{z}\end{aligned}$$

(b)

$$E = \frac{1}{2} m v^2 + V(r)$$

$$\begin{aligned}v^2 &= \vec{v} \cdot \vec{v} \\ &= (\dot{r} \hat{r} + r \dot{\theta} \hat{\theta}) \cdot (\dot{r} \hat{r} + r \dot{\theta} \hat{\theta}) \\ &= \dot{r}^2 + r^2 \dot{\theta}^2\end{aligned}$$

Thus,

$$E = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \dot{\theta}^2 + V(r)$$

7.

$$(a) \quad (8, \pi/4, \pi/6) \equiv (r, \theta, \phi)$$

$$x = r \sin \theta \cos \phi = 4 \sqrt{\frac{3}{2}}$$

$$y = r \sin \theta \sin \phi = 4/\sqrt{2}$$

$$z = r \cos \theta = 8/\sqrt{2}$$

$$(b) \quad r = \sqrt{x^2 + y^2} = 4\sqrt{3}$$

$$\tan \theta = y/x = \frac{6}{2\sqrt{3}} = \sqrt{3}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

$$z = -4$$

$$\begin{aligned} &2 \times 2^4 \\ &2 \times 2 \times 12 \\ &2 \times 2 \times 2 \times 3 \\ &\quad \times 2 \\ &4 \times 4 \times 3 \end{aligned}$$

$$\text{Thus, } (2\sqrt{3}, \pi/3, -4) \rightarrow (4\sqrt{3}, \frac{\pi}{3}, -4)$$

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Solution to Practice problems

8. (a) $\hat{r} = \sin\theta \cos\phi \hat{x} + \sin\theta \sin\phi \hat{y} + \cos\theta \hat{z}$

$$\hat{\theta} = \cos\theta \cos\phi \hat{x} + \cos\theta \sin\phi \hat{y} - \sin\theta \hat{z}$$

$$\hat{\phi} = -\sin\phi \hat{x} + \cos\phi \hat{y}$$

(b) $\hat{x} = \sin\theta \cos\phi \hat{r} + \cos\theta \cos\phi \hat{\theta} - \sin\phi \hat{\phi}$

$$\hat{y} = \sin\theta \sin\phi \hat{r} + \cos\theta \sin\phi \hat{\theta} + \cos\phi \hat{\phi}$$

$$\hat{z} = \cos\theta \hat{r} - \sin\theta \hat{\theta}$$

9.

It can be proved by showing that

$$\hat{r} \cdot \hat{\theta} = 0$$

$$\hat{r} \cdot \hat{\phi} = 0$$

$$\hat{\theta} \cdot \hat{\phi} = 0$$

10.

$$(a) \quad (r, \theta, z) = \left(3, \frac{\pi}{3}, -4\right)$$

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$z = z$$

$$x = \frac{3}{2}, \quad y = \frac{3\sqrt{3}}{2}, \quad z = -4$$

$$\left(3, \frac{\pi}{3}, -4\right) \longrightarrow \left(\frac{3}{2}, \frac{3\sqrt{3}}{2}, -4\right)$$

$$(b) \quad (x, y, z) = (-2, 2, 3)$$

$$r = \sqrt{x^2 + y^2} = 2\sqrt{2}$$

$$\tan \theta = \frac{y}{x} \Rightarrow \tan \theta = -1$$

$$\Rightarrow \theta = 3\frac{\pi}{4}$$

$$z = 3$$

Thus,

$$(-2, 2, 3) \longrightarrow \left(2\sqrt{2}, 3\frac{\pi}{4}, 3\right) //$$